Preface

Recognition, by scientists and nonscientists alike, of the oceans' important role in our climate has grown enormously in the past 25 years. In part, this growth was driven by the World Ocean Circulation Experiment (WOCE), the planning for which started in the 1980s and on which the first edition of this book, published in 2001, was focused. Although the issues relating to climate variability and change were well known in 2001, they occupy a far more central position in oceanography and climate science in 2013. The increased focus on climate change and its socioeconomic consequences has arisen from the accumulation of unambiguous evidence of substantial changes in the climate system associated with human activities. The oceans are central to these changes as we document increases in the ocean heat and carbon content, sea-level rise, Arctic sea-ice loss, ocean acidification, and more. These changes are so important that they provide the impetus for many present-day ocean research programs. Hence it is no accident that a number of authors of the present volume also played key roles in the Fifth Assessment Report (AR5) of the Intergovernmental Panel on Climate Change (IPCC) completed in October 2013. The ocean science community is increasingly being asked to provide expertise to help chart a path toward sustainability in a world where significant change is inevitable. In parallel with this increased recognition, our understanding of the oceans and our capability to observe and model them has improved enormously. WOCE was a major driver of this improvement and progress has continued to be made by WOCE's successor projects in the World Climate Research Programme, by initiatives of the Global Climate Observing System and by many national programs.

We have been struck, as we trust our readers will be, by the enormous progress in ocean observations and modeling that has been made since the publication of the first edition of *Ocean Circulation and Climate*. In 2001, the first floats were deployed in the Argo profiling float array. Argo has since become a mainstay of our *in situ* ocean-observing system. In 2004, routine monitoring of the Atlantic Meridional Overturning Circulation started and continues to this day. Since the beginning of the twenty-first century, our observational capability, data delivery, and quality control systems have advanced to the point where ocean indices and gridded fields can be presented on a weekly or monthly basis, not only at the ocean surface but also for the subsurface ocean. We have a new and rigorous equation of the state of seawater and a redefinition of salinity published in 2010. We can now investigate interannual, and to some degree decadal, variability and change of sea level using more than 20 years of satellite altimetry measurements.

The first goal of WOCE was to develop ocean models for use in climate research. The significant improvement of coupled atmosphere-ocean general circulation models has in part been underpinned by progress made during and since WOCE, but has also been aided by the relentless increase in computing power. Models of the global ocean commonly, though not yet routinely, include mesoscale features (eddies, fronts, boundary currents) that were extremely rare in models circa 2001. Consequently, simulations, particularly those assimilating observations, play a central role in interpreting the current state of the ocean and in aiding the planning for new measurement programs. Progress in computer power has allowed not only more realistic representations of ocean physics but also the inclusion of ocean biogeochemistry. These models also no longer rely on artificial flux corrections to prevent model drift.

Copyright © 2013. Elsevier Science & Technology. All rights reserved.

Ocean Circulation and Climate : A 21st Century Perspective, Elsevier Science & Technology, 2013. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/utas/detail.action?docID=1495673.
Created from utas on 2021-08-05 22:46:52.

A deeper conceptual understanding of the role that the oceans play in climate has been prompted by the many improvements to the observational record and enhanced modeling capabilities. The particular role that the oceans play in long-term climate variations, trends, and in the predictability of these changes is greatly aided by idealized conceptual models, with significant progress made along these lines over the past decade.

This book aims at presenting ocean science and its relevance to climate more than a decade after the first edition of *Ocean Circulation and Climate*. When we were approached by Elsevier to consider editing a second edition, we were initially reticent, remembering the magnitude of the work involved in producing the first edition. However, our peers and younger colleagues reassured us of the value of the first edition as a unique resource documenting the state of knowledge of the oceans' role in climate following WOCE; a project in which we were all variously involved. We were encouraged to develop the second edition presenting "A 21st Century Perspective."

The achievements attained by the end of WOCE provided an almost entirely physically focused "snapshot" of the ocean as documented in the first edition. However, since 2001, there has been progress on a broader front. In part, this is due to the significant lengthening of the record of changing ocean properties with the WOCE era observations providing a "state of the ocean in the 1990s" benchmark. The greater maturity and comprehensiveness of the network of ocean observations was founded during WOCE and was subsequently stimulated by the 1999 and 2009 OceanObs conferences, with the second of these having a more interdisciplinary focus. For this new edition of the book, we aimed to have a strong involvement of scientists whose careers developed during and after WOCE. Our authors include a great proportion of younger scientists from the post-WOCE generation.

This second edition of *Ocean Circulation and Climate* leads the reader through the important areas of progress since the first edition. It is composed of six parts each prefaced by a short introduction. The reader is initiated in the ocean's role in climate by Part I. In Parts II – IV we discuss developments in ocean observations, processes in the ocean and elements of the global-scale circulation. Modeling of the ocean and the coupled climate system and its changes are discussed in Parts V and VI. While the book focuses on ocean physics, we have also included related paleoclimatic findings and descriptions of biogeochemical processes, marine ecosystems, and the carbon cycle. There are inevitable overlaps between chapters, and cross-references are inserted where appropriate. Production of the book ran in parallel with the preparation of the IPCC AR5 for which this book provides an ocean-focused complement addressing the underpinning scientific issues.

Despite the enormous progress made during and since the WOCE era, many challenges remain. Perhaps first and foremost is maintaining the *in situ* and satellite observational capability and filling the remaining gaps to provide a comprehensive knowledge and understanding of how and why the oceans are changing. Development of ocean and climate models will require continued and strengthened partnerships between ocean, atmospheric, and cryospheric modelers and with the observational community. The ocean–cryosphere interaction is an important and urgent challenge.

There are also new opportunities, particularly as the length of observational time series increases and as new and innovative techniques become established. For example, this edition has little focus on the various gravity missions that are beginning to bring new insights into ocean circulation and its variability. These missions are likely to have a much bigger impact on a subsequent edition of this book. More powerful computers and improved ocean model representation will allow greater resolution of ocean features and greater coupling of physical, cryospheric, biogeochemical, and ecosystem elements of the ocean.

Copyright © 2013. Elsevier Science & Technology. All rights reserved.

Ocean Circulation and Climate : A 21st Century Perspective, Elsevier Science & Technology, 2013. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/utas/detail.action?docID=1495673.
Created from utas on 2021-08-05 22:46:52.

With continued enthusiasm within the ocean science community to address the challenging issues of ocean climate, we are confident that the future for understanding the ocean circulation, its interaction with the atmosphere and the cryosphere, and the related biogeochemical and ecosystem sciences is bright.

Gerold Siedler
Helmholtz Centre for Ocean Research
Kiel, Germany

Stephen M. Griffies
NOAA Geophysical Fluid Dynamics Laboratory
Princeton, USA

John Gould
National Oceanography Centre
Southampton, UK

John A. Church
Centre for Australian Weather and Climate Research
A Partnership between CSIRO and the Bureau of
Meteorology, Hobart, Australia

Copyright © 2013. Elsevier Science & Technology. All rights reserved.

Ocean Circulation and Climate : A 21st Century Perspective, Elsevier Science & Technology, 2013. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/utas/detail.action?docID=1495673.
Created from utas on 2021-08-05 22:46:52.

This page intentionally left blank

Copyright © 2013. Elsevier Science & Technology. All rights reserved.

Ocean Circulation and Climate : A 21st Century Perspective, Elsevier Science & Technology, 2013. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/utas/detail.action?docID=1495673.
Created from utas on 2021-08-05 22:46:52.